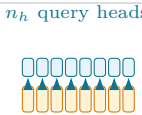
one query token, per layer: → reads shared K/V sets → $C \approx 2\,L\,T\,n_{kv}\,d_h\,b$ (b: bytes per element)



			cache size, one scale (less sharing ⇒ more expressive)
MHA	own K, V for every query head	n_h KV groups	$C_{\text{MHA}} \approx 2LTn_hd_hb$ — 100%
GQA	1 K, V per group of heads	n_{kv} groups, $1 < n_{kv} < n_h$	$C \approx 2LTn_{kv}d_hb$ — ≈ 25%
MQA	all heads share one K, V	1 KV group	$C \approx 2LTd_hb$ — ≈ 3.1%
MLA	one latent per token, rebuild K, V	1 latent + position	$C \approx LT(d_c + d_R)b$ — ≈ 7.0%

Schematic: 8 query heads drawn. Percentages use $n_h = 32$, $d_h = 128$, $n_{kv} = 8$; MLA latent $d_c = 512$ plus position branch $d_R = 64$.
More sharing ⇒ smaller cache, cheaper decode; more independent K/V subspaces ⇒ more expressive.